For instance, the equation $x^3 - 2 = 0$ has only one irrational root, namely $\sqrt[3]{2}$. Of course, the other two roots are imaginary numbers (What are they?).

**Example 3.9**

Find a polynomial equation of minimum degree with rational coefficients, having $2 - \sqrt{3}$ as a root.

**Solution**

Since $2 - \sqrt{3}$ is a root and the coefficients are rational numbers, $2 + \sqrt{3}$ is also a root. A required polynomial equation is given by

$$x^2 - (\text{Sum of the roots})\ x + \text{Product of the roots} = 0$$

and hence

$$x^2 - 4x + 1 = 0$$

is a required equation. ■

**Note**

We note that the term "rational coefficients" is very important; otherwise, $x - (2 - \sqrt{3}) = 0$ will be a polynomial equation which has $2 - \sqrt{3}$ as a root but not $2 + \sqrt{3}$. We state the following result without proof.

**Theorem 3.4**

Let $p$ and $q$ be rational numbers so that $\sqrt{p}$ and $\sqrt{q}$ are irrational numbers; further let one of $\sqrt{p}$ and $\sqrt{q}$ be not a rational multiple of the other. If $\sqrt{p} + \sqrt{q}$ is a root of a polynomial equation with rational coefficients, then $\sqrt{p} - \sqrt{q}, -\sqrt{p} + \sqrt{q}$, and $-\sqrt{p} - \sqrt{q}$ are also roots of the same polynomial equation.

**Example 3.10**

Form a polynomial equation with integer coefficients with $\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ as a root.

**Solution**

Since $\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ is a root, $x - \sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ is a factor. To remove the outermost square root, we take $x + \sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ as another factor and find their product

$$\left(x + \sqrt{\frac{\sqrt{2}}{\sqrt{3}}}\right)\left(x - \sqrt{\frac{\sqrt{2}}{\sqrt{3}}}\right) = x^2 - \frac{\sqrt{2}}{\sqrt{3}}.$$

Still we didn't achieve our goal. So we include another factor $x^2 + \frac{\sqrt{2}}{\sqrt{3}}$ and get the product

$$\left(x^2 - \frac{\sqrt{2}}{\sqrt{3}}\right)\left(x^2 + \frac{\sqrt{2}}{\sqrt{3}}\right) = x^4 - \frac{2}{3}.$$

So, $3x^4 - 2 = 0$ is a required polynomial equation with the integer coefficients. ■

Now we identify the nature of roots of the given equation without solving the equation. The idea comes from the negativity, equal to 0 and positivity of $\Delta = b^2 - 4ac$.



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